Project B - System Architecture

1. Overview

Project B is a Django monolith backed by a MySQL database. It is deployed on two virtual machines behind a hardware load balancer. One codebase serves the web interface for supervisors and the JSON API used by the handheld scanner application. Sessions are stored in Redis.

2. Deployment diagram

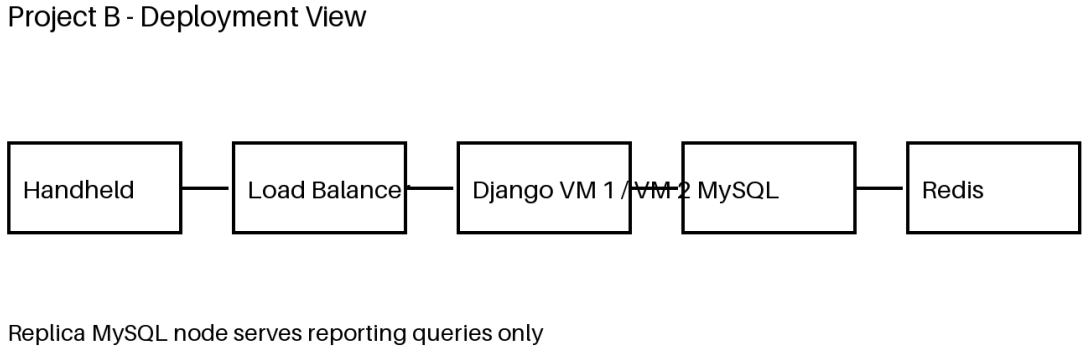

Figure 1: Project B deployment view.

3. Data model

Every scan creates an immutable row in movement; stock_level is derived and can be rebuilt from movements. Errors are corrected by compensating movements instead of edits.

Table	Purpose	Approx rows
item	Master list of stock items	60,000
location	Warehouses and bays	900
stock_level	Derived current quantity	95,000
movement	Immutable record of every scan	41,000,000
adjustment	Supervisor-approved corrections	12,500

4. Offline handling

Handhelds queue scans locally during network loss and upload them in order afterwards. A client-generated identifier on each scan prevents duplicates on retry.